

# ANN5: Bidirectional Associative Memory (BAM) – Detailed Code & Viva Explanation

## Cell 1: Import Library

**Code:**

```
import numpy as np
```

**Explanation (what to say in viva):**

Here I am importing numpy for numerical operations.

It is used for matrix multiplication and array handling.

BAM heavily relies on matrix operations.

## Cell 2: Define Input and Output Patterns

**Code:**

```
X = np.array([[1, -1], [-1, 1]])  
Y = np.array([[1, 1], [-1, -1]])
```

**Explanation (what to say in viva):**

X represents input patterns.

Y represents corresponding output patterns.

BAM stores associations between X and Y.

Each input pattern is linked with an output pattern.

## Cell 3: Compute Weight Matrix

**Code:**

```
W = np.dot(X.T, Y)
```

**Explanation (what to say in viva):**

Here we compute the weight matrix.

X transpose is multiplied with Y.

This follows Hebbian learning rule.

It stores relationships between input and output patterns.

## Cell 4: Activation Function

**Code:**

```
def sign(x):  
    return np.where(x>=0, 1, -1)
```

**Explanation (what to say in viva):**

This is the sign activation function.

If value is positive → 1, else → -1.

It converts continuous values into bipolar output.

BAM uses bipolar representation.

## Cell 5: Forward Recall

### Code:

```
y_pred = sign(np.dot(X, W))
```

### Explanation (what to say in viva):

Here we calculate forward recall.

Input X is multiplied with weight matrix W.

Sign function is applied to get output.

This retrieves stored output patterns.

## Cell 6: Backward Recall

### Code:

```
x_pred = sign(np.dot(Y, W.T))
```

### Explanation (what to say in viva):

Here we calculate backward recall.

Output Y is multiplied with transpose of W.

This retrieves original input patterns.

This shows bidirectional property of BAM.

## Cell 7: Print Output

### Code:

```
print(y_pred)  
print(x_pred)
```

### Explanation (what to say in viva):

This prints predicted output and input.

We verify if recall is correct.

If output matches original, model is working properly.